

Sources

AIS GMVO portal
daily discharge
and stage
2008–2023

Processing and screening

Parsing and harmonisation
gap filling ≤ 6 d
reservoir stage: ≤ 15 d

discharge

Discharge-only quality grading
completeness + 16 flags
→ grades A–F

Released files

discharge.nc
water_level.nc

year_grades.csv
year_flags.csv

MERIT Hydro DEM
AIS GMVO + Hydroweb:
gauge coordinates
Hydroweb: reference areas

Watershed delineation,
manual boundary verification

boundaries.gpkg

Gridded forcing
MSWEP, ERA5-Land,
GPCP, GLEAM4
+ boundaries.gpkg

Basin-averaged
forcing aggregation

forcing.nc

HydroATLAS v1.0
+ boundaries.gpkg

Attribute extraction

attributes.csv

Discharge, forcing,
catchment area

Signature computation

signatures.csv